

姓名：

(號)

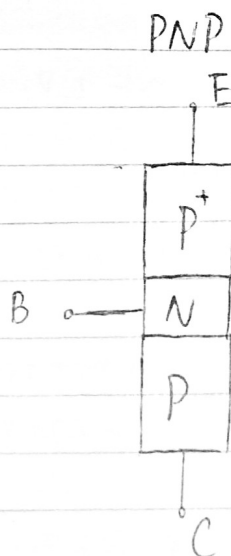
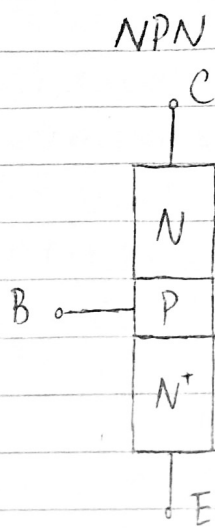
成績

年級：

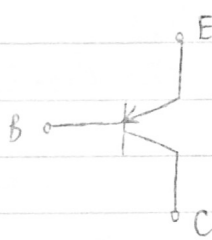
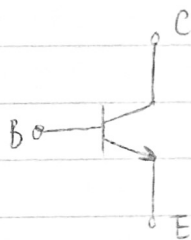
科目：

作業 1

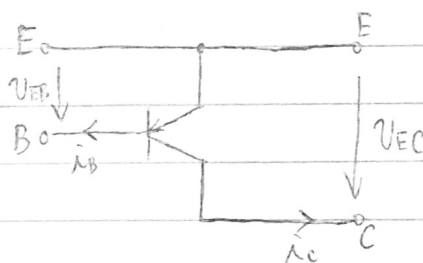
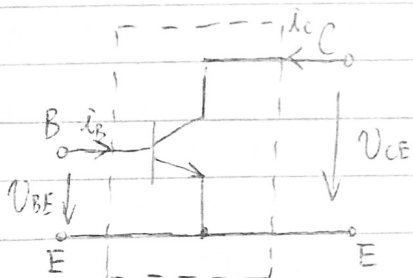
結構



電路符號



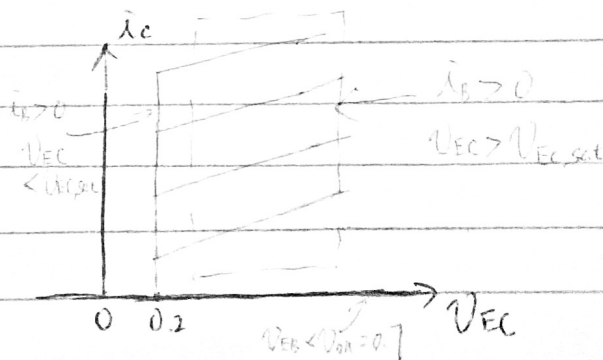
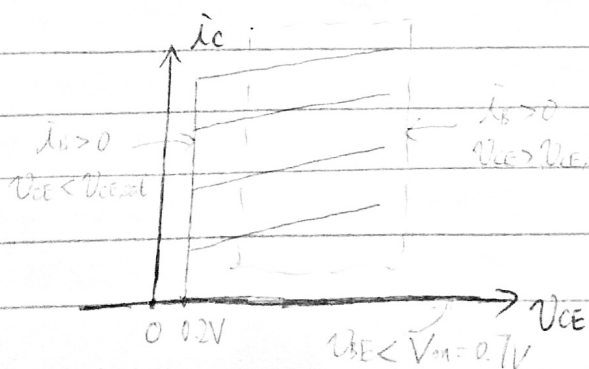
二端口定義



有源區元件約束

$$\begin{aligned} i_B &= I_{BS0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \\ i_C &= \beta_n i_B \left(1 + \frac{V_{CE}}{V_A} \right) \\ &= \beta_n I_{BS0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \left(1 + \frac{V_{CE}}{V_A} \right) \end{aligned}$$

$$\begin{aligned} i_B &= I_{BS0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \\ i_C &= \beta_p i_B \left(1 + \frac{V_{EC}}{V_A} \right) \\ &= \beta_p I_{BS0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right) \left(1 + \frac{V_{EC}}{V_A} \right) \end{aligned}$$



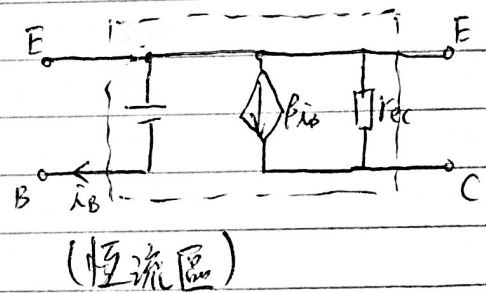
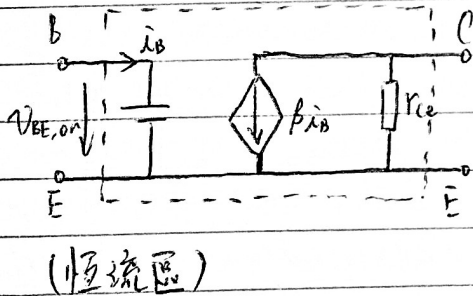
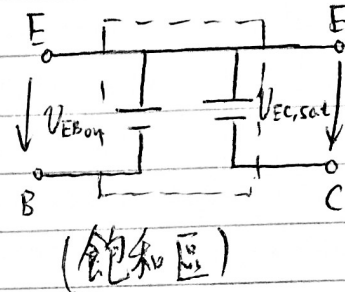
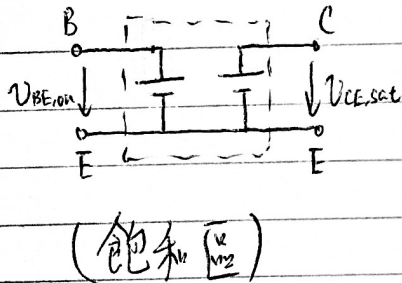
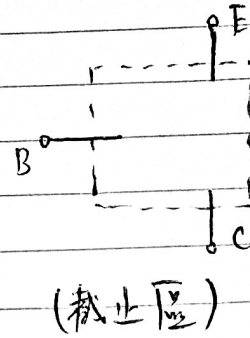
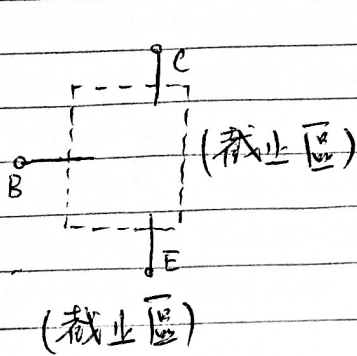
姓名：

(號)

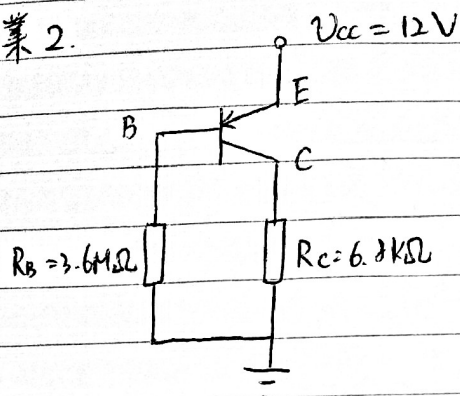
成績

年級：

科目：



作業 2.



$$V_{EB} = 0.7V$$

$$\bar{i}_B = \frac{V_{CC} - V_{EB}}{R_B} = \frac{11.3V}{3.6M\Omega} = 3.139\mu A$$

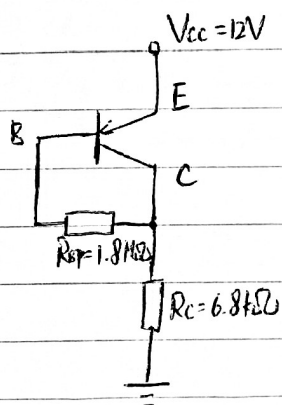
$$\begin{aligned} V_C &= \bar{i}_C R_C = \beta \bar{i}_B R_C = (150 \sim 400) \times 3.139\mu A \times 6.8k\Omega \\ &= (150 \sim 400) \times 21.34mV \\ &= 3.20 \sim 8.54V \end{aligned}$$

$$V_{EC} = V_{CC} - V_C = 12V - (3.20 \sim 8.54V)$$

$$= (3.46V \sim 8.80V) > 0.2V = V_{EC, set}$$

所以 BJT 工作在恒流區

$$\bar{i}_C = 470.85\mu A \sim 1.255mA$$



$$V_{EB} = 0.7V$$

$$V_C = (\bar{i}_B + \bar{i}_C) R_C = (\beta + 1) \bar{i}_B R_C$$

$$\bar{i}_B = \frac{V_{CC} - V_{EB} - V_C}{R_{BP}} = \frac{11.3V - 6.8k\Omega \cdot (\beta + 1) \bar{i}_B}{1.8M\Omega}$$

$$\bar{i}_B \left(1 + \frac{34}{9} \times 10^{-3} (\beta + 1) \right) = 6.28\mu A$$

$$\bar{i}_B = \frac{6.28\mu A}{1 + \frac{34}{9} \times 10^{-3} \times (151 \sim 401)} = 2.50\mu A \sim 3.99\mu A$$

$$V_{EC} = V_{CC} - V_C = 12V - (\beta + 1) \bar{i}_B R_C$$

$$= 12V - \frac{6.28\mu A}{\frac{1}{\beta + 1} + \frac{34}{9} \times 10^{-3}} \times 6.8k\Omega = 5.19V \sim 7.89V > 0.2V$$

所以 BJT 工作在恒流區

$$\bar{i}_C = \beta \bar{i}_B = \frac{6.28\mu A}{\frac{1}{\beta} + \frac{34}{9} \times 10^{-3} (1 + \frac{1}{\beta})} = 599.8\mu A \sim 998.9\mu A$$

培道中學校

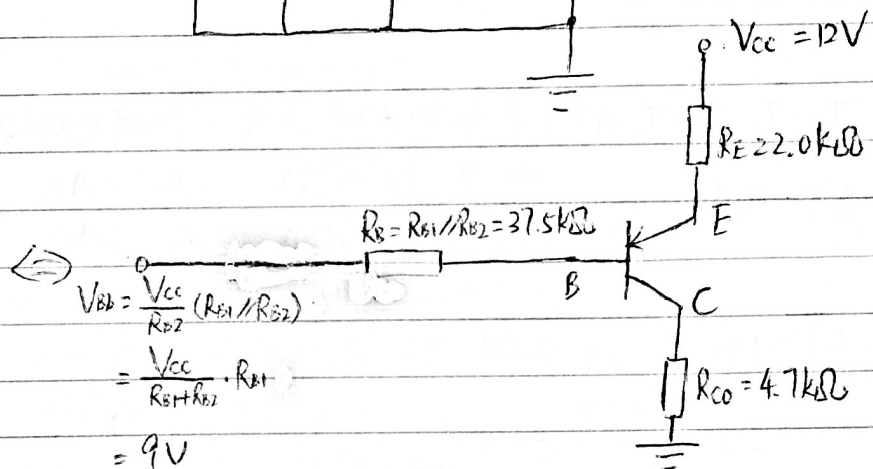
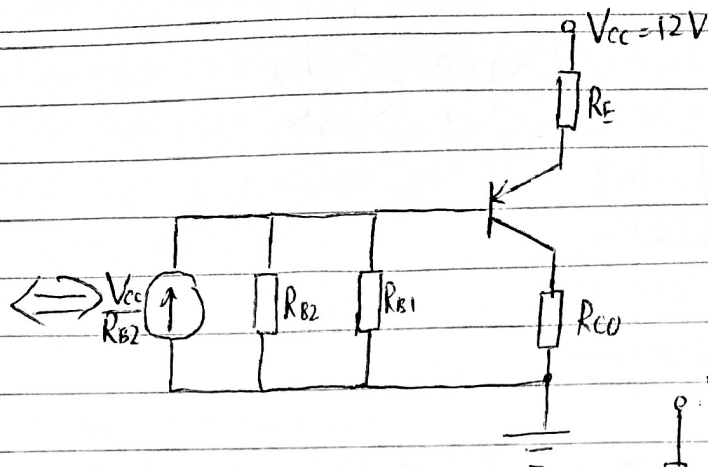
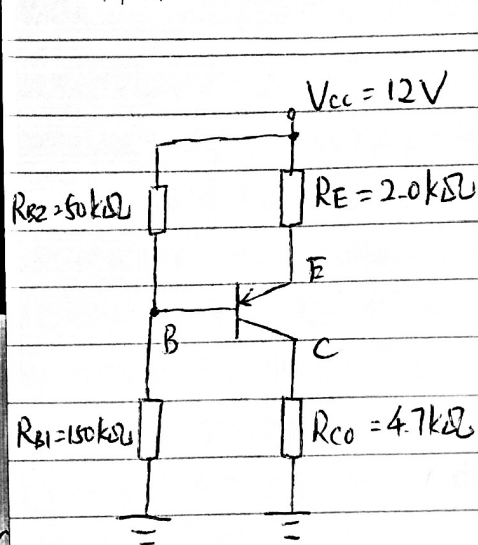
姓名：

(號)

成績

年級：

科目：



$$V_{CC} - V_{BB} = (\beta + 1) \bar{I}_B R_E + \bar{I}_B R_B + V_{EB}$$

$$2.3V = \bar{I}_B ((\beta + 1) R_E + R_B) = \bar{I}_B [(\beta + 1) 2.0k\Omega + 37.5k\Omega]$$

$$\bar{I}_B = \frac{2.3V}{37.5k\Omega + (\beta + 1) \times 2k\Omega} = 2.74\mu A \sim 6.77\mu A$$

$$\bar{I}_C = \frac{2.3V}{\frac{37.5k\Omega}{\beta} + (1 + \frac{1}{\beta}) \times 2k\Omega} = 1.01mA \sim 1.09mA$$

$$V_{EC} = V_{CC} - (\beta + 1) \bar{I}_B R_E - \beta \bar{I}_B R_C$$

$$= V_{BB} + \bar{I}_B R_E + V_{EB} - \beta \bar{I}_B R_C$$

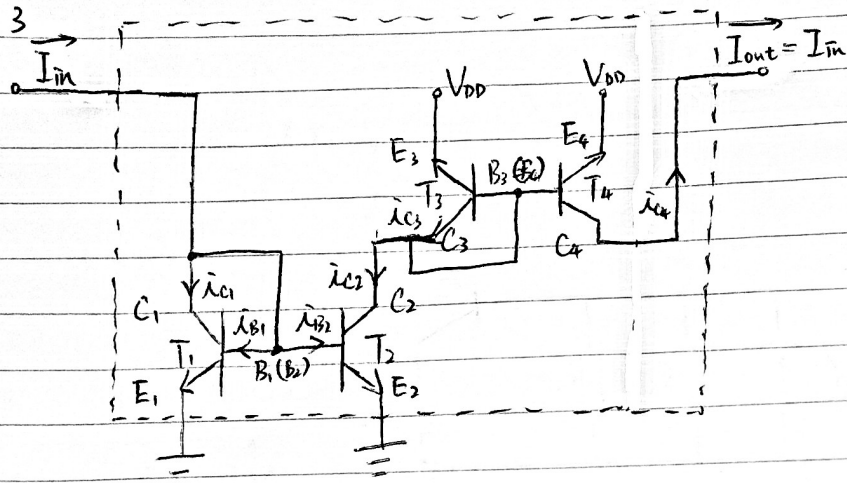
$$= 9.7V + \frac{2.3V}{\frac{37.5k\Omega}{\beta} + (\beta + 1) \times 2k\Omega} \times (2k\Omega - 4.7k\Omega \beta)$$

$$= 9.7V + \frac{4.6V}{19.75 + \beta} - \frac{10.81V}{\frac{37.5}{\beta} + 2} > 0.2V$$

所以BJT工作區性線區)

$\bar{I}_C = 1.01mA \sim 1.09mA$ (最穩定)

作業 3



前提: T_1, T_2, T_3, T_4 的工藝參量相差無幾 (忽略厄利效應)

$$\text{在 } T_1, T_2 \text{ 中, } \frac{\lambda_{C1}}{\lambda_{C2}} = \frac{\beta A_{J1} J_{S0} (e^{\frac{V_{D1, E1}}{V_T}} - 1)}{\beta A_{J2} J_{S0} (e^{\frac{V_{D2, E2}}{V_T}} - 1)} = \frac{\beta_1}{\beta_2} \cdot \frac{A_{J1}}{A_{J2}} = 1$$

$$\text{同理可知 } \lambda_{B1} = \lambda_{B2} = \frac{\lambda_{C1}}{\beta}$$

$$\text{所以 } I_{in} = \lambda_{C1} + \lambda_{B1} + \lambda_{B2} = (1 + \frac{2}{\beta}) \lambda_{C1} = (1 + \frac{2}{\beta}) \lambda_{C2}$$

在 T_3, T_4 中, 也有 $\lambda_{B3} = \lambda_{B4}$ (因為 $V_{E3, B3} = V_{DD} - V_{B3} = V_{DD} - V_{B4} = V_{E4, B4}$)

$$\text{所以 } \lambda_{C2} = \lambda_{C3} + \lambda_{B3} + \lambda_{B4} = (1 + \frac{2}{\beta}) \lambda_{C4} = (1 + \frac{2}{\beta}) I_{out}$$

$$\text{因此有 } I_{in} = (1 + \frac{2}{\beta})^2 I_{out} \approx I_{out}$$

$$\text{不妨取 } \beta = 150 \sim 400, \text{ 則 } \frac{\Delta I_{in}}{I_{in}} = \frac{I_{out} - I_{in}}{I_{in}} = (1 + \frac{2}{\beta})^2 - 1 = 1.0025\% \sim 2.6844\%$$

(誤差範圍可被允許)

姓名：

(號)

成績

年級：

科目：

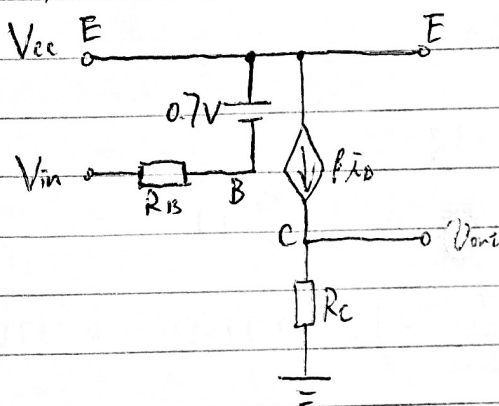
作業 4

(1) 當 BE 反偏, BC 反偏時, $\bar{i}_B = \bar{i}_C = 0$

$$V_{EB} = V_{CC} - V_{in} < V_{on} \Leftrightarrow V_{in} > V_{CC} - V_{on} = V_{CC} - 0.7V$$

$$V_{out} = 0V$$

(2) 當 BE 正偏, BC 反偏時,



$$\bar{i}_B = \frac{V_{CC} - V_{in} - V_{on}}{R_B}$$

$$V_{out} = \bar{i}_C R_C = \beta \frac{V_{CC} - V_{in} - V_{on}}{R_B} R_C$$

$$= \beta (V_{CC} - 0.7V) \frac{R_C}{R_B} - \beta \frac{R_C}{R_B} V_{in}$$

$$\text{此時 } V_{in} < V_{CC} - 0.7V$$

$$\text{且 } V_{EC} = V_{CC} - V_{out} > V_{EC,sat}$$

$$\Leftrightarrow V_{out} < V_{CC} - V_{EC,sat}$$

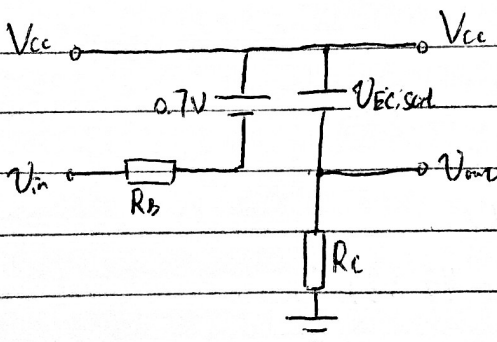
$$\Leftrightarrow \beta (V_{CC} - 0.7V) \frac{R_C}{R_B} - \beta \frac{R_C}{R_B} V_{in} < V_{CC} - V_{EC,sat}$$

$$\Leftrightarrow \beta \frac{R_C}{R_B} V_{in} > \beta (V_{CC} - 0.7V) \frac{R_C}{R_B} - (V_{CC} - V_{EC,sat})$$

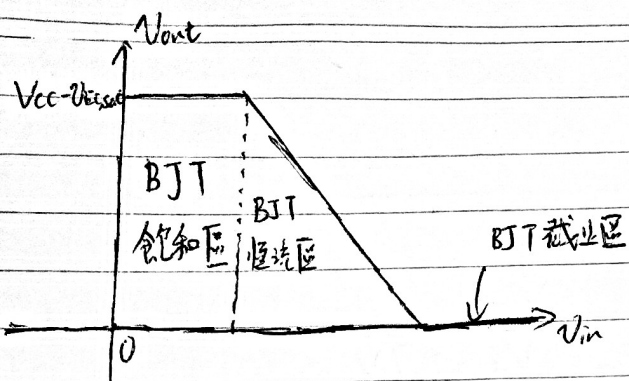
$$\Leftrightarrow V_{in} > V_{CC} - 0.7V - \frac{V_{CC} - V_{EC,sat}}{\beta \frac{R_C}{R_B}}$$

$$\Leftrightarrow V_{CC} - 0.7V - \frac{V_{CC} - V_{EC,sat}}{\beta \frac{R_C}{R_B}} < V_{in} < V_{CC} - 0.7V$$

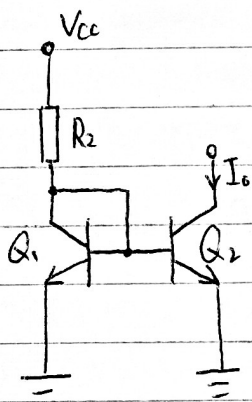
(3) 當 BE 正偏, BC 正偏時,



$$V_{out} = V_{CC} - V_{EC,sat}$$



習題 5



首先, $I_{mA} = I_C = I_{CS0} (e^{\frac{V_{BE}}{V_T}} - 1) = I_{CS0} (e^{\frac{700}{25}} - 1)$

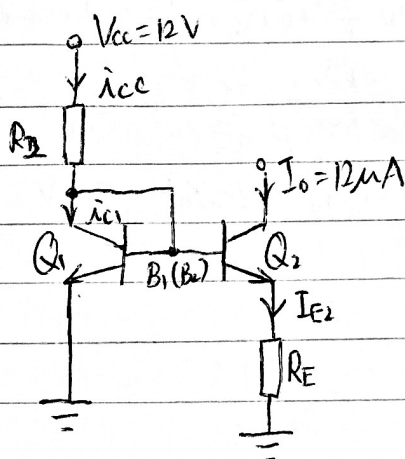
$$\Leftrightarrow I_{CS0} = 1.482 \text{ fA}$$

假設 Q_1, Q_2 的工藝參量一致

現有 $I_0 = I_{C1} = 12 \mu\text{A} = 1.482 \text{ fA} \cdot (e^{\frac{V_{BE}}{V_T}} - 1)$

$$\Rightarrow V_{BE} = V_T \ln \left(\frac{I_{C1}}{I_{CS0}} + 1 \right) = 587.29 \text{ mV} = 0.587 \text{ V}$$

$$R_2 = \frac{V_{CC} - V_{BE}}{I_{C1} \times (1 + \frac{1}{\beta})} \approx \frac{V_{CC} - V_{BE}}{I_{C1}} = 0.95 \text{ M}\Omega$$



$$I_{E2} = I_0 (1 + \frac{1}{\beta}) \approx I_0$$

$$V_{BE2} = 0.587 \text{ V (上題分析結果)}$$

$$I_{CC} = I_{C1} (1 + \frac{1}{\beta}) + \frac{I_0}{\beta} \approx I_{C1}$$

$$I_{C1} = I_{CS0} (e^{\frac{V_{BE2} + I_{E2} R_E}{V_T}} - 1)$$

$$V_{BE2} + I_{E2} R_E + I_{CC} R_2 = V_{CC}$$

不妨取 $I_{CC} = 1 \text{ mA}$, 此時 $V_{BE2} + I_{E2} R_E = 0.7 \text{ V}$

$$R_E = \frac{0.7 \text{ V} - 0.587 \text{ V}}{12 \mu\text{A}} = 99.38 \text{ k}\Omega$$

$$R_2 = \frac{V_{CC} - V_{BE2}}{I_{CC}} = 11.3 \text{ k}\Omega$$

姓名：

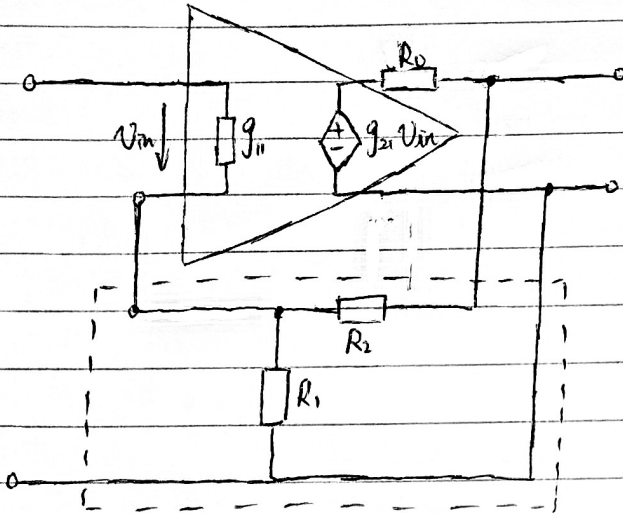
(號)

成績

年級：

科目：

作業 6



$$g_A = \begin{pmatrix} g_{11} & 0 \\ A_{v0} & R_0 \end{pmatrix}$$

$$h_A = \frac{1}{g_{11} R_0} \begin{pmatrix} R_0 & 0 \\ -A_{v0} & g_{11} \end{pmatrix} = \begin{pmatrix} R_{11} & 0 \\ -A_{v0} R_{11} g_0 & g_0 \end{pmatrix}$$

$$h_F = \begin{pmatrix} R_1 \parallel R_2 & \frac{R_1}{R_1 + R_2} \\ -\frac{R_1}{R_1 + R_2} & \frac{1}{R_1 + R_2} \end{pmatrix}$$

$$h_{AF} = h_A + h_F = \begin{pmatrix} R_{11} + \frac{R_1 R_2}{R_1 + R_2} & \frac{R_1}{R_1 + R_2} \\ -A_{v0} R_{11} g_0 - \frac{R_1}{R_1 + R_2} & g_0 + \frac{1}{R_1 + R_2} \end{pmatrix}$$

$$g_{AF} = \begin{pmatrix} g_0 + \frac{1}{R_1 + R_2} & -\frac{R_1}{R_1 + R_2} \\ A_{v0} R_{11} g_0 + \frac{R_1}{R_1 + R_2} & R_{11} + \frac{R_1 R_2}{R_1 + R_2} \end{pmatrix}$$

$$\left(R_{11} + \frac{R_1 R_2}{R_1 + R_2} \right) \times \left(g_0 + \frac{1}{R_1 + R_2} \right) + \left(A_{v0} R_{11} g_0 + \frac{R_1}{R_1 + R_2} \right) \cdot \frac{R_1}{R_1 + R_2}$$